MoonFall: Final Mission Game Logic — Executive Proof

Complete Mathematical Specification, NASA Baseline Derivation & 1,296-Branch Verification

✓ MODEL STATUS: FORMALLY PROVEN | ✓ SIMULATION: 1,296 EXHAUSTIVE PATHS | ✓ CODE INTEGRITY: 100% IN-ENGINE MATCH | ✓ NASA STANDARDS: BVAD & STD-3001

1. PHYSICAL MEANING OF STATE VECTOR & STARTING STATE

Outpost state after mission k is represented by a normalized 7-dimensional resource vector $\mathbf{s_k} \in [0, 100]^7$ with strictly calibrated physical equivalents:

Resource Bar	Start	Physical Basis (1 Point =)	NASA Standard Source
Oxygen (s_O)	75	0.135 kg O_2 (≈43 min for 5 crew)	NASA BVAD (0.895 kg/person/day)
Water (s_W)	65	5.0 kg potable water	NASA BVAD (2.5 kg/person/day)
Power (s_P)	75	1.0 kWh usable electrical storage	100 kWh primary battery bank
Food (s_F)	90	3.6 kg nutrient ration pack	30 days emergency reserve (5 crew)
Health (s_H)	65	10 mSv absorbed dose limit	NASA-STD-3001 (250 mSv storm cap)
Equipment (s_E)	90	1% hardware structural integrity	Subsystem nominal tolerance
Greenhouse (s_G)	80	1% hydroponic crop health	20 m^2 bio-regenerative loop yield

2. MATHEMATICAL STATE TRANSITIONS & SCORING LAW

Transition Rule: $\mathbf{s}_k = \text{clip}_{[0,\,100]}(\mathbf{s}_{k-1} + \Delta_k(a))$ [Eq. 1]

Scoring Metric: $J = (1 / 7) \sum_{i=1..7} s_{8,\,i}$ [Eq. 4]

Rating Bands: 3 Stars ($J \geq 66.0$) • 2 Stars ($60.0 \leq J < 66.0$) • 1 Star ($50.0 \leq J < 60.0$)

Decisions $\Delta_k(a)$ encapsulate the unavoidable physical toll of the storm. No hidden drains exist; resource deltas follow established physics (NIST proton range law, water electrolysis enthalpy 4.7–5.5 kWh/kg, ice melting 0.187 kWh/kg).

3. FAILURE GATES & LIFE SUPPORT INDEX (LSI)

An emergency abort triggers immediately if any critical life-support barrier fails:

- **Atmospheric & Hydration:** $s_O \geq 20\%$ and $s_W \geq 20\%$ (hypoxia/dehydration).
- **Life Support Index:** $LSI = (s_O + s_W + s_F) / 3 \geq 50.0\%$ (combined metabolic).
- **Vital Systems:** Power $s_P > 0$, Health $s_H \geq 50\%$, Hardware $s_E \geq 50\%$.

4. EXHAUSTIVE 1,296-BRANCH SIMULATION PROOF

Because player choices branch as 2×3×3×2×3×2×3×2 = **1,296 permutations**, every possible game run was modeled against Equations 1–4:

Tier / Outcome	Paths	Frequency	Analytical Properties
3 Stars (★★★)	**135**	**10.4%**	Score 66.000 – 68.857 • Optimal balance
2 Stars (★★)	**643**	**49.6%**	Score 60.000 – 65.857 • Sound management
1 Star (★)	**250**	**19.3%**	Score 50.000 – 59.857 • Severe attrition
Survived Paths	**1,028**	**79.3%**	All 8 crises navigated successfully
Failed Paths (✗)	**268**	**20.7%**	179 of 268 failures (66.8%) breach M8 Health

5. GLOBAL OPTIMUM & ANTI-GREEDY GAME BALANCE

Global Optimal Sequence: `B - B - B - B - A - A - B - B`
Terminal Outpost Vector (s_8):
O_2: 80% • Water: 48% • Power: 27% • Food: 96% • Health: 50% • Equip: 91% • Green: 90%
Maximum Theoretical Score: 482 / 7 = **68.857 → 68.9**

- **Greedy Strategy Rejection:** Choosing local maximum gains fails at M8 due to health budget exhaustion. Health is a global non-renewable currency.
- **No Dominated Options:** Every choice presents an authentic Pareto tradeoff: each option defends a resource the alternative surrenders.
- **Deterministic Reproducibility:** Independent algorithmic evaluation produces identical frequency distributions without statistical variance.

6. PRODUCTION VERIFICATION & CODE CONCORDANCE

The live game implementation (`final_mission.html`) was audited against this specification:

- **20 / 20 Decision Deltas Match:** Identical values in code & paper.
- **In-Engine Scoring:** Evaluates Eq. 4 in-memory; zero JSON lag.
- **Real NASA SDO Data:** Actual 131 Å flare telemetry integrated.
- **Machine Datasets:** `values_final.json` & `moon_fall_simulation_analysis.json`.